

Step 2: Create a Security Group

A security group acts as a virtual firewall to control the traffic for its associated instances. To use a security group, you add the inbound rules to control incoming traffic to the instance, and outbound rules to control the outgoing traffic from your instance. To associate a security group with an instance, you specify the security group when you launch the instance. If you add and remove rules from the security group, we apply those changes to the instances associated with the security group automatically.

Your VPC comes with a *default security group*. Any instance not associated with another security group during launch is associated with the default security group. In this exercise, you'll create a new security group, `WebServerSG`, and specify this security group when you launch an instance into your VPC.

Topics

- [Rules for the WebServerSG Security Group](#)
- [Creating Your WebServerSG Security Group](#)

Rules for the WebServerSG Security Group

The following table describes the inbound and outbound rules for the `WebServerSG` security group. You'll add the inbound rules yourself. The outbound rule is a default rule that allows all outbound communication to anywhere — you do not need to add this rule yourself.

Inbound			
Source IP	Protocol	Port Range	Comments
0.0.0.0/0	TCP	80	Allows inbound HTTP access from anywhere.
0.0.0.0/0	TCP	443	Allows inbound HTTPS access from anywhere.
Public IP address range of your home network	TCP	22	Allows inbound SSH access from your home network to a Linux/UNIX instance.
Public IP address range of your home network	TCP	3389	Allows inbound RDP access from your home network to a Windows

			instance.
Outbound			
Destination IP	Protocol	Port Range	Comments
0.0.0.0/0	All	All	The default outbound rule that allows all outbound communication.

Creating Your WebServerSG Security Group

You can create your security group using the Amazon VPC console.

To create the WebServerSG security group and add rules

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
2. In the navigation pane, choose **Security Groups**.
3. Choose **Create Security Group**.
4. In the **Group name** field, enter `WebServerSG` as the name of the security group, and provide a description. You can optionally use the **Name tag** field to create a tag for the security group with a key of `Name` and a value that you specify.
5. Select the ID of your VPC from the **VPC** menu, and then choose **Yes, Create**.
6. Select the `WebServerSG` security group that you just created (you can view its name in the **Group Name** column).
7. On the **Inbound Rules** tab, choose **Edit** and add rules for inbound traffic as follows, and then choose **Save** when you're done:
 - a. Select **HTTP** from the **Type** list, and enter `0.0.0.0/0` in the **Source** field.
 - b. Choose **Add another rule**, then select **HTTPS** from the **Type** list, and enter `0.0.0.0/0` in the **Source** field.
 - c. Choose **Add another rule**. If you're launching a Linux instance, select **SSH** from the **Type** list, or if you're launching a Windows instance, select **RDP** from the **Type** list. Enter your network's public IP address range in the **Source** field. If you don't know this address range, you can use `0.0.0.0/0` for this exercise.

Caution

.....

If you use 0.0.0.0/0, you enable all IP addresses to access your instance using SSH or RDP. This is acceptable for the short exercise, but it's unsafe for production environments. In production, you'll authorize only a specific IP address or range of addresses to access your instance.

Summary

Inbound Rules

Outbound Rules

Tags

Cancel

Save

Type	Protocol	Port Range	Source	Remove
HTTP (80)	TCP (6)	80	0.0.0.0/0 i	✖
HTTPS (443)	TCP (6)	443	0.0.0.0/0 i	✖
SSH (22)	TCP (6)	22	192.0.2.0/24 i	✖
RDP (3389)	TCP (6)	3389	192.0.2.0/24 i	✖

Add another rule